

- Q.31 Write short note on heating with humidification. (CO3)
- Q.32 Discuss disadvantages of air refrigeration over vapour compression system (Co1)
- Q.33 Show DBT, WBT and relative humidity lines on psychrometric chart. (CO7)
- Q.34 Enlist various methods of refrigeration. (CO1)
- Q.35 Write short note on spray pond. (CO4)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain central air conditioning system in detail with neat sketch. (CO7)
- Q.37 What is domestic electrolux refrigeration system? Explain its working principle with neat sketch. (CO5)
- Q.38 Explain vapour compression refrigeration cycle in detail with neat sketch. (CO2)

No. of Printed Pages : 4
Roll No.

181752/171752
/121752/031752

5th Sem / Mech Subject:- Refrigeration and Air Conditioning

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Brine solution is the mixture of (CO1)
a) Salt and Water b) Nitrogen and Salt
c) Water and Milk d) None of these
- Q.2 COP of a heat pump is always (CO2)
a) Less than one b) Equal to one
c) Greater than one d) None of these
- Q.3 Heart of the refrigeration cycle is (CO2)
a) Condenser b) Compressor
c) Expansion valve d) Evaporator
- Q.4 One ton of refrigeration is equal to (CO4)
a) 210 KJ/min b) 110KJ/min
c) 320 KJ/min d) 450 KJ/min
- Q.5 Which type of system is best for cooling of a cinema hall (CO4)
a) Window air conditioner
b) Automobile air conditioner
c) Central air conditioner
d) Split air conditioner

- Q.6 Dry ice is also called as (CO4)
 a) CO_2 b) NO_2
 c) Solid CO_2 d) CO
- Q.7 In a refrigeration cycle, heat is rejected by the refrigerant in a (CO2)
 a) Compressor b) Condenser
 c) Evaporator d) Expansion valve
- Q.8 Air conditioning is the process of control of (CO7)
 a) Temperature b) Humidity
 c) Cleanliness d) All of these
- Q.9 The relative humidity of the air should not above _____ In summer for comfort cooling. (CO4)
 a) 30% b) 40%
 c) 50% d) 60%
- Q.10 The Capillary tube is a kind of (CO5)
 a) Compression device b) Expansion device
 c) Filter d) None of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define refrigeration. (CO1)
- Q.12 What is the principle of vapour compression refrigeration cycle. (CO1)
- Q.13 Write the function of suction line in simple vapour compression. (CO3)
- Q.14 What is the use of flash chamber? (CO5)

(2) 181752/171752
 /121752/031752

- Q.15 What is other name of domestic electrolux refrigeration system? (CO1)
- Q.16 Name two types of rotary compressors. (CO5)
- Q.17 Define humidification. (CO6)
- Q.18 What is sensible heat factor? (CO5)
- Q.19 Define COP. (CO3)
- Q.20 Define Auto defrosting. (CO5)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define one ton of refrigeration. (CO2)
- Q.22 Write five properties of R-22. (CO3)
- Q.23 Explain overload protector. (CO5)
- Q.24 Describe the function of refrigerants. (CO8)
- Q.25 How to select refrigerants for Vapour compression refrigeration system. (CO8)
- Q.26 Write short note on inverter technology. (CO5)
- Q.27 What are the disadvantages of solar power refrigeration system over vapour compression system? (CO5)
- Q.28 What is hermetically sealed compressor. (CO4)
- Q.29 Explain effect of sub cooling in vapour compression refrigeration system. (CO2)
- Q.30 State some applications of refrigeration and air conditioning. (CO5)

(3) 181752/171752
 /121752/031752